

## Trainings ▾

### General Information

#### Gear Driven

Some K38 and K50 marine series engines are equipped with two sea water pumps. The pumps are mounted on supports on the front gear cover. The supports for the left bank and the right bank are different. The sea water pumps are the same for both the left and right banks. The sea water pump is a gear driven pump, that is driven by the camshaft gear.

The sea water pump on K38 and K50 marine series engines contains a paddle type impeller that allows the pump to function in either a **clockwise** or **counterclockwise** rotation. When viewed from the front of the engine, the sea water pump on the right bank rotates in a **counterclockwise** direction. The sea water pump on the left bank rotates in a **clockwise** direction. The inlet for the sea water pump is the port side nearest to the front gear cover.

The bearings in the sea water pumps on K38 and K50 marine series engines are lubricated by engine oil through drillings in the front gear cover and the pump supports.

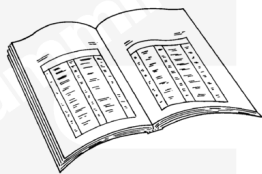
The sea water pumps on K38 and K50 marine series engines are limited to engine speeds of 2000 rpm or less. Cummins Inc., does **not** provide sea water pump gears for engine speeds in excess of 2000 rpm. The sea water pumps mounted on K38 and K50 marine series engines are Gilmec, Model 250/700, manufactured by:

- Gilbert Gilkes & Gordon Ltd.
- Kendal, Cumbria
- England
- LA9 7B2

### Preparatory Steps

#### Gilkes

- Isolate the water supply to the sea water pump.
- Disconnect the piping from the sea water pump.

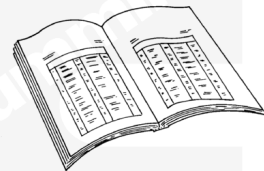


ck800wa

## Jabsco®

- Isolate the water supply to the sea water pump.
- Disconnect the piping from the sea water pump.

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ck800wa

## Remove

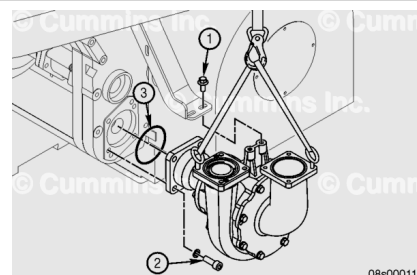
### Gilkes

#### **⚠ WARNING ⚠**

**This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.**

#### **⚠ CAUTION ⚠**

**Do not support the weight of the sea water pump on its shaft end. Damage to the shaft can result.**



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Without the mounting adapter:

Place a strap around the pump to support the weight of the pump during removal.

Remove the two supporting bracket cap screws (1).

Remove the four mounting cap screws and locking washers (2).

Work the pump loose from the gear housing and remove the pump from the engine.

Discard the o-ring (3).

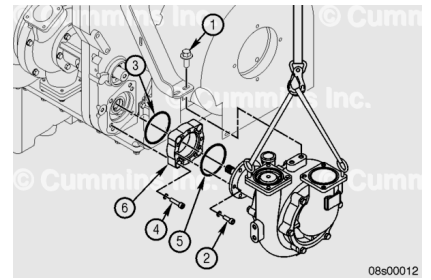
## Jabsco®

### ⚠ WARNING ⚠

**This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.**

### ⚠ CAUTION ⚠

**Do not support the weight of the sea water pump on its shaft end. Damage to the shaft can result.**



With the mounting adapter:

Place a strap around the pump to support the weight of the pump during removal.

Remove the two supporting bracket cap screws (1).

Remove the six mounting cap screws and washers (2).

Work the pump loose from the gear housing adaptor and remove the pump from the engine.

Discard the o-ring (5).

Remove the four adaptor mounting cap screws and locking washers (4).

Remove the adaptor (6) from the front gear cover. Discard the o-ring (3).

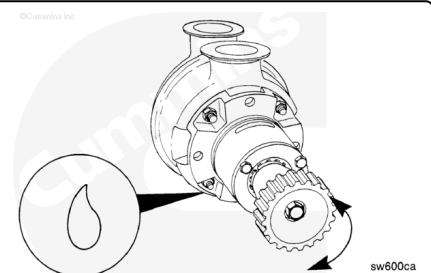
## Disassemble

### Gear Driven

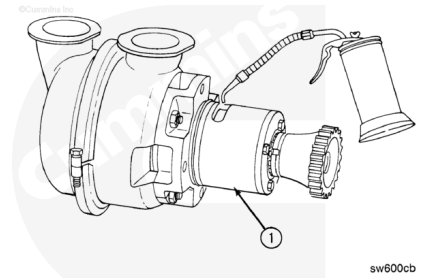
Rotate the shaft to check for damaged bearings or parts that are rubbing.

If the pump does **not** turn correctly, refer to Refer to Procedure 008-074 in Section 8.

(/qs3/pubsys2/xml/en/procedures/28/28-008-074.html)



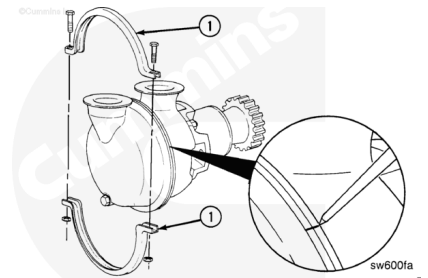
While turning the shaft, put clean 15W-40 engine oil in the oil passage to lubricate the bearings.



**Note :** This procedure can also apply to check or replace the seal when the pump is installed on the engine. If the work is being performed with the pump installed, the hoses for the pump must be removed.

Remove the V-band clamp (1).

Use a scribe or a center punch. Put a mark on both ends of the water pump bodies so that they can be assembled correctly.



**Note :** The body is a tight fit. If necessary, use a wooden wedge to aid in the separation of the parts. Put the wedge in the space between the inlet and outlet ports.

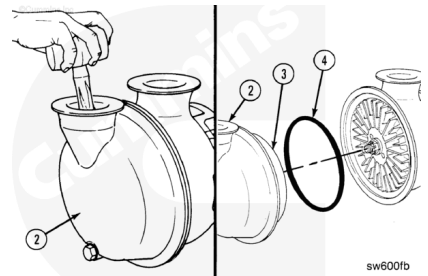
Insert a wooden hammer handle through the part until the handle is touching the end of the shaft.

Pry on the handle.

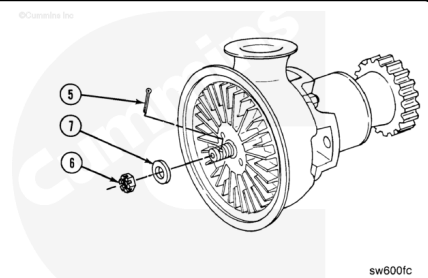
Remove the nondrive end of the body (2).

**Note :** The spacer ring (3) can be a tight fit in the nondrive end of the body. It is not necessary to remove the spacer ring if it is secure in the body (2).

Remove the o-ring seal (4).

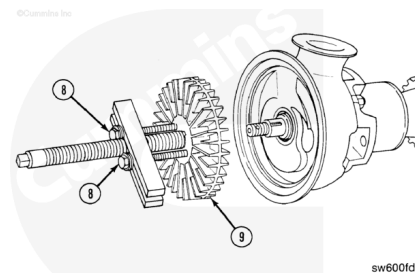


Remove the cotter pin (5), slotted nut (6), and washer (7).



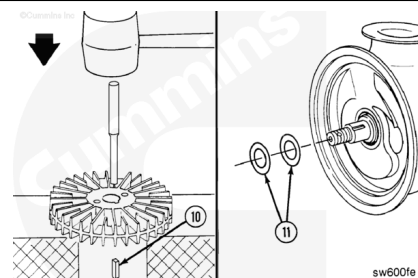
Use standard puller ST-647, or equivalent, and two 8 - 1.25 x 70 mm capscrews (8).

Remove the impeller (9).



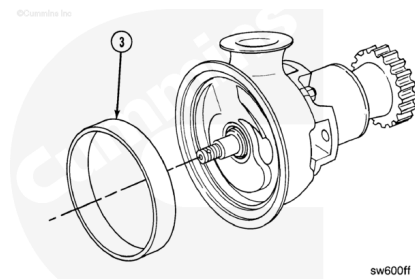
Use a drift and a mallet. Remove the key (10) from the impeller.

Remove the shims (11). Keep the shims together for future assembly. The thickness of the shims is critical for the performance of the pump.



**Note :** The spacer ring can be a tight fit in the body. If necessary, use two pry bars to remove the ring.

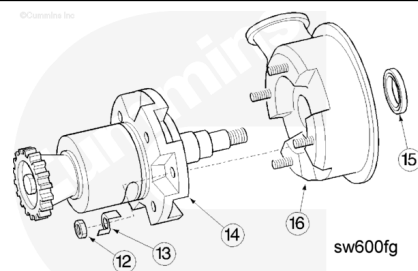
Remove the spacer ring (3).



Bend the lockplates. Remove the four nuts (12) and the lockplates (13).

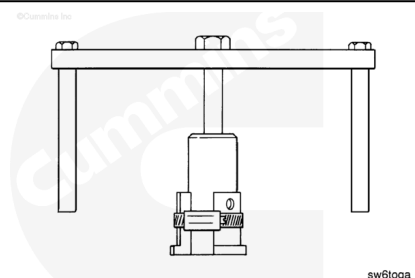
**Note :** It will be necessary to use a mallet to separate the parts because of the tight fit between the shaft and the water seal and spacer.

Remove the drive end of the body (14), and spacer (15), from the bearing housing and shaft assembly (16).

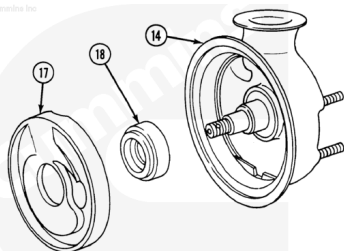


**Note :** The port plate can be a tight fit in the body after many hours of operation. If necessary, use the special tool available from Gilkes™.

**Note :** This tool can also be used to remove the port plate from the nondrive end of the body.



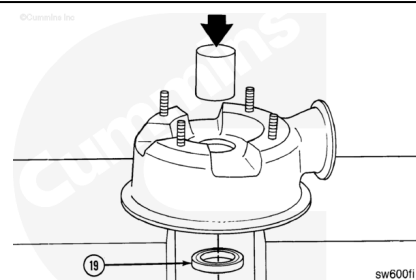
Remove the port plate (17) and mechanical seal (18) from the drive end of the pump body (14).



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Use a press and an appropriate mandrel to remove the contact ring (19) from the body.

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sw600fi

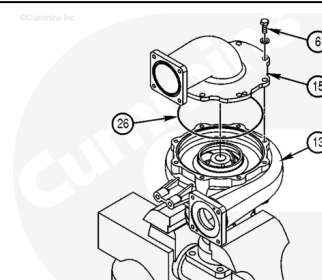
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Remove the eight suction bend mounting capscrews (6).

Remove the suction bend (15) from the pump case (13).

Remove the o-ring (26).

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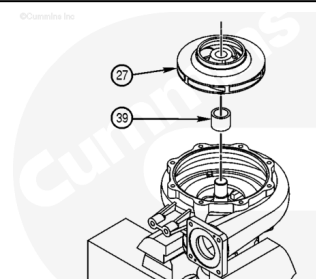
**Note :** The main impeller uses a left hand thread.

Restrain the shaft spline.

Use a strap wrench to unscrew the main impeller (27).

Remove the spacer (39).

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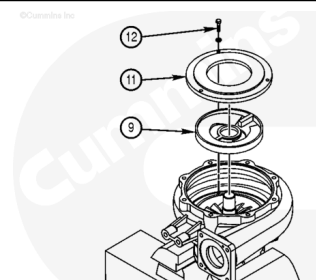


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Loosen and remove the three screws (12) that hold the wear ring (11) into position.

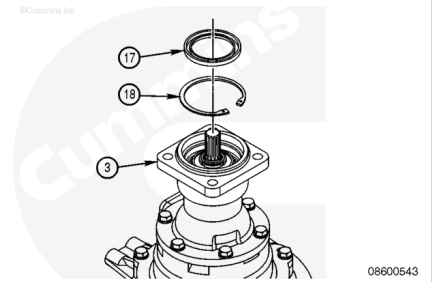
Remove the wear ring and the visible port plate (9).

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Rotate the assembly to remove the lip seal (17) and bevelled circlip (18) from the drive end of the bearing housing (3).

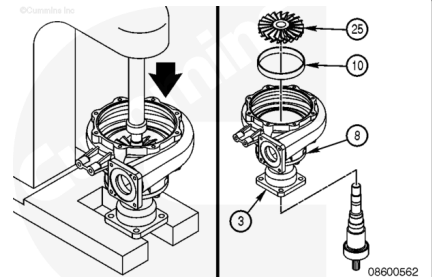


Stand the assembly in a press, supporting it on the square drive end flange. Press down on the shaft end.

As the shaft is pressed through the intermediate body (8) and the bearing housing (3), the priming impeller (25) will be released.

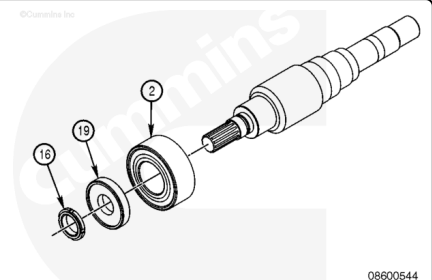
Take care that the shaft does **not** fall.

Remove the distance ring (10).



Unscrew and remove the locknut (16).

Press the bearing (2) and seal collar (19) off the shaft. This can be tight as it is secured with Loctite® 574 threadlocker.

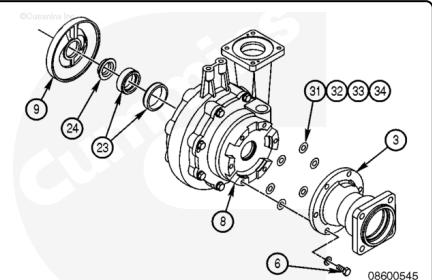


Remove the six screws (6) holding the bearing housing (3) to the intermediate body (8). Make sure to catch the shims (31), (32), (33), and (34).

Press the second port plate (9) out of the intermediate body (8).

Remove the seal spacer (24).

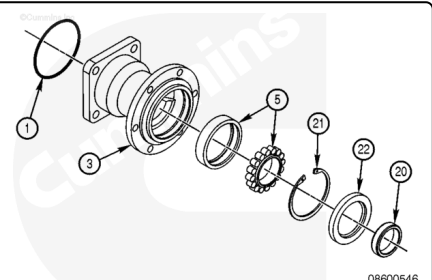
Remove the loose rotating half of the mechanical seal (23) and press out the stationary half.



Remove the seal collar (20), the lip seal (22), and remove the circlip (21) from the bearing housing. The inner race of the roller bearing (5) will drop out.

Press out the bearing outer race.

Remove the o-ring (1).



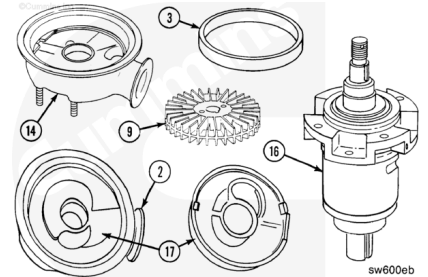
# Clean and Inspect for Reuse

## Gear Driven

### **⚠ WARNING ⚠**

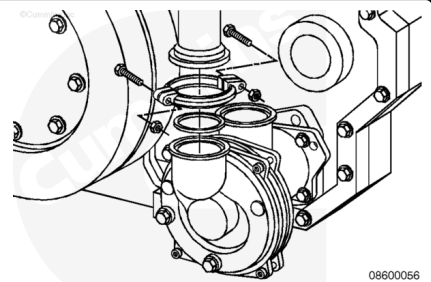
**When using detergents for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the risk of personal injury.**

Use a 5-percent detergent solution to clean the parts. It is important to eliminate all of the grease and dirt from the surfaces that contact the contact ring and the mechanical seal.



Check the impeller (1) and port plate for damage from cavitation.

If necessary, replace the parts.

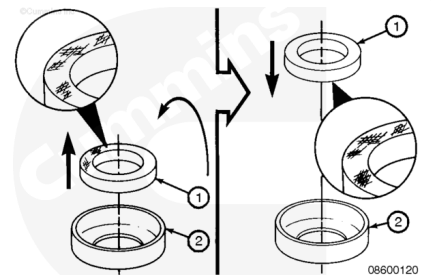


Gilkes™ recommends the replacement of the mechanical seal **only** when the contact ring is replaced at the same time.

**Note :** The following is **only** a temporary repair for an emergency situation. Cummins Inc., does **not** recommend the use of this procedure for a permanent repair. The correct repair procedures must be followed as soon as possible.

For an emergency repair, the contact ring (1) can be reversed so that the side of the ring that is **not** worn touches the mechanical seal.

Reverse the ring and install it in the rubber boot (2).



## Gilkes

### **⚠ WARNING ⚠**



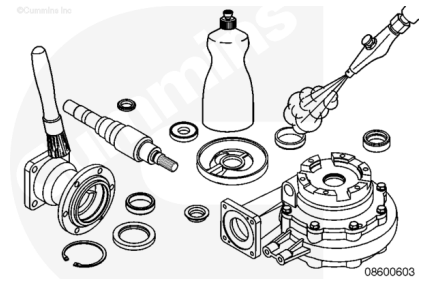
When using detergents for cleaning, follow the manufacturers recommendations for use. To reduce the possibility of personal injury, wear goggles and protective clothing.

**⚠ WARNING ⚠**

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

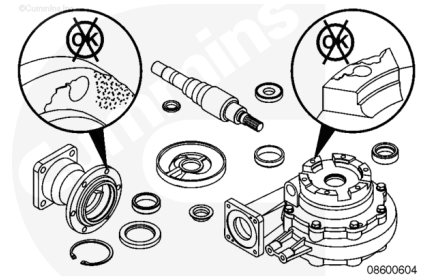
Use a 5-percent detergent solution to clean the parts.

Dry with compressed air.



Inspect all sea water pump components for evidence of cracks, pitting, and any obvious damage.

Replace any damaged parts as necessary.



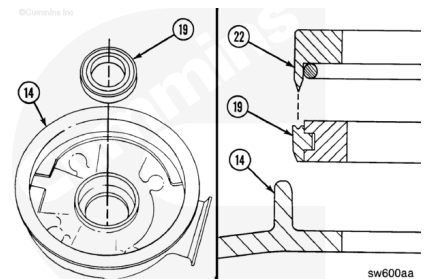
## Assemble

### Gear Driven

Use a 5-percent detergent solution. Lubricate the rubber boot. Do **not** use grease.

Use the special tool (22) that is available from Gilkes™ or install the contact ring (19) by hand in the drive end of the body (14).

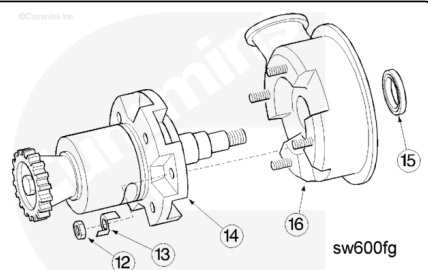
Make sure the ring touches the bottom of the bore in the body.



Use a 5-percent detergent solution. Lubricate the inside diameter of the contact ring.

Install the drive end of the body (14) on the bearing housing and shaft assembly (16).

Install the lockplates (13) and the nuts (12).

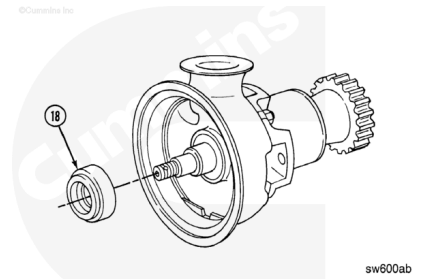


**Torque Value:** 34 n•m [ 25 ft-lb ]

Bend the lockplate tabs over the nuts and the edge of the housing.

**⚠ WARNING ⚠**

**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**



**⚠ CAUTION ⚠**

**Do not touch the sealing surface of the seal. The oil from your hands can cause the seal to fail.**

Use solvent to clean the shaft and the contact ring.

Use a 5-percent detergent solution to lubricate the shaft.

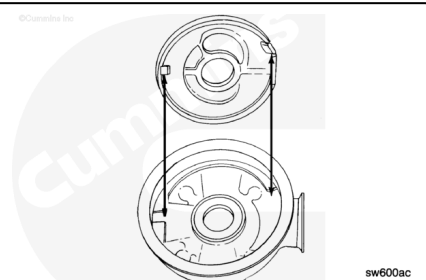
**Note :** The largest diameter of the seal must touch the contact ring.

Install the mechanical seal (18).

Push the seal until it touches the contact ring.

**⚠ CAUTION ⚠**

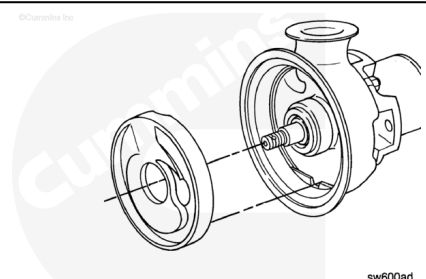
**The lugs on the back of the port plate must enter the slots in the body.**



**Note :** The port plates are identical and interchangeable.

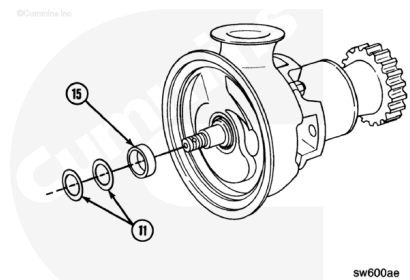
Install the drive end of the port plate.

The port plate will be approximately 13 mm [0.500 in] below the chamfer when the plate is installed correctly.



Install the spacer (15).

Install the shims (11). The shims **must** touch the drive end of the port plate.

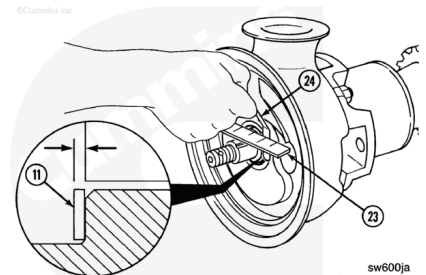


**Note :** The thickness of the shims is critical to the performance of the pump.

Use a straight edge (23) or ring that will touch the step in the shaft.

Use a feeler gauge (24), to measure the distance from the step on the shaft and the shims (straight edge) to the port plate.

This distance is the impeller side clearance.



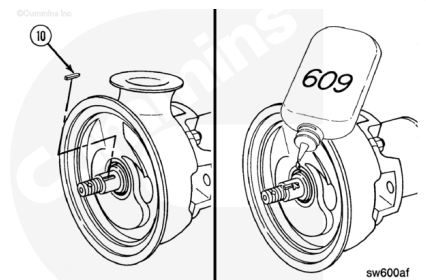
| Sea Water Pump Impeller Side Clearance |     |       |
|----------------------------------------|-----|-------|
| mm                                     |     | in    |
| 0.18                                   | MIN | 0.007 |
| 0.23                                   | MAX | 0.009 |

If the side clearance is **not** within specifications, remove or install shims (11) until the side clearance is correct.

Make sure the contact ring, seal, and port plate are seated properly.

Install the shaft key (10).

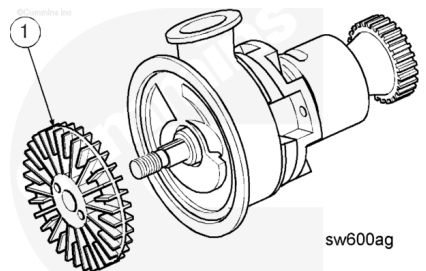
Apply Loctite® 609, or equivalent, to the shaft and key.



**⚠ CAUTION ⚠**

**Do not use a hammer and mandrel to install the impeller. The bearings in the pump will be damaged.**

**Note :** Make sure the puller holes in the impeller are positioned toward the nondrive end of the pump.



Align the slot in the impeller with the key. Install the impeller on the shaft.

Use a mallet with a soft nose. Tap the impeller (1) **only** until it is straight on the shaft.

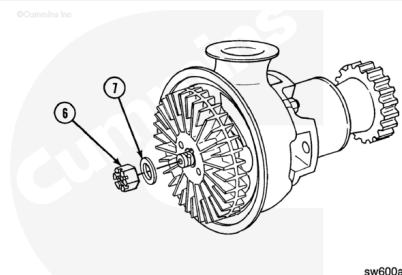
**⚠ CAUTION ⚠**

**Do not use too much grease or antiseize compound. If the grease touches the seal, the seal will fail.**

Use water pump grease or antiseize compound. Apply a light layer to the threads on the shaft.

Install the washer (7) and the nut (6).

Use the nut to push the impeller on the shaft. The impeller will stop when it contacts the shims.



**⚠ CAUTION ⚠**

**Use this illustration to identify the torque specification for the nut being installed. If the older, smaller nut is torqued to the higher specification, the nut will fail.**

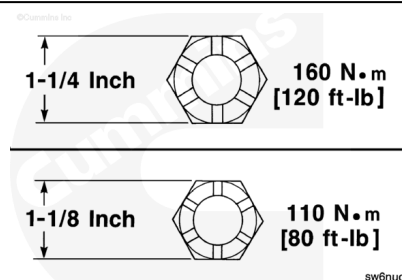
The nut requires a 0.05 mm [1-1/4 in] wrench:

**Torque Value:** 160 n•m [120 ft-lb]

The nut requires a 0.044 mm [1-1/8 in] wrench:

**Torque Value:** 110 n•m [80 ft-lb]

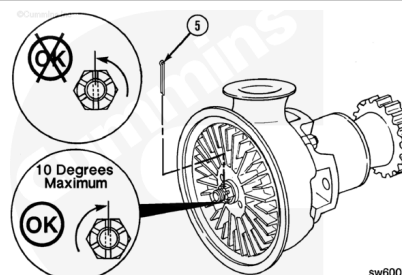
The larger nut is recommended.



**⚠ CAUTION ⚠**

**Do not loosen the nut to align the slots in the nut with the hole in the shaft. Tighten the nut no more than 10 degrees to align the holes. If the slots and holes are not in alignment, change the washer or reduce the thickness of the washer slightly so that a slot and the hole are in alignment at the proper torque.**

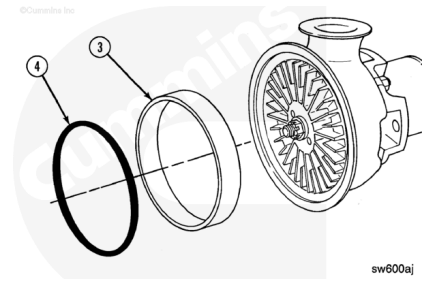
Install the cotter pin (5). Bend the ends of the pin over the nut.



Install the o-ring seal (4) on the spacer ring (3).

Install the spacer ring and o-ring seal. Push the o-ring seal under the lip for the V-Band clamp.

**Note :** If the nondrive end of the port plate was removed, install the plate in the body so that the lugs on the plate engage the slots in the body.



Install the nondrive end of the body (2).

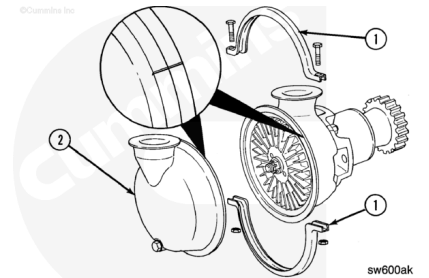
Align the marks scribed on the bodies during disassembly. The center of the inlet and outlet ports **must** be in alignment within 3 degrees.

Install the V-Band clamp (1).

Tighten the clamp.

**Torque Value:** 20 n•m [ 15 ft-lb ]

Use a plastic hammer. Tap around the V-Band clamp completely. Torque the clamp fasteners again. Continue this step until the fasteners remain at the correct torque after tapping the clamp.



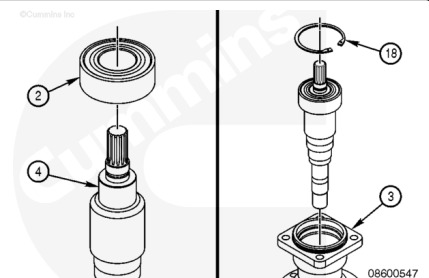
## Gilkes

Mechanical seals, lip seals, o-rings, spring washers, and bearings are **not** serviceable, and **must always** be replaced.

**Note :** Whenever Loctite® products are mentioned, all surfaces **must** be clean and grease free in accordance with Loctite® brand recommendations.

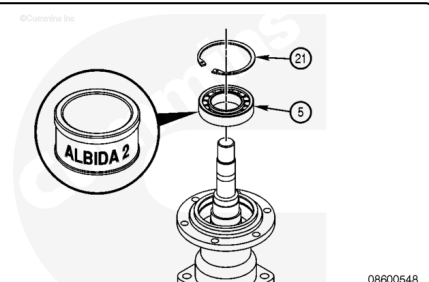
Press the bearing (2) onto the shaft (4).

Press the shaft and bearing assembly into the bearing housing (3) with a bevelled circlip (18).



Press the inner race of the roller bearing (5) into position on the shaft. Fully pack this bearing with Shell® ALBIDA "2" grease.

Secure the bearing with the circlip (21).

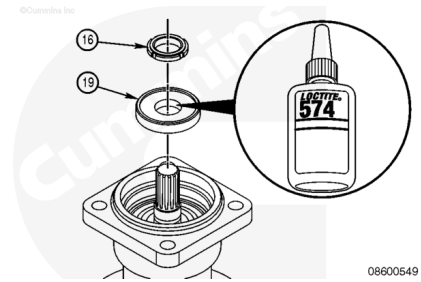


Apply Loctite® 574 threadlocker to the face in contact with the shaft.

Fit the seal collar (19) over the shaft.

Secure with a locknut (16).

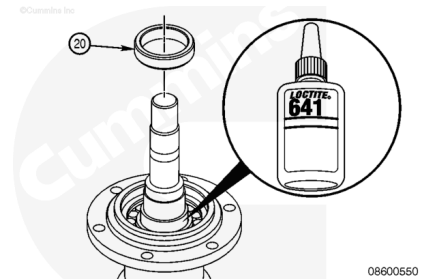
**Torque Value:** 50 n•m [ 37 ft-lb ]



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Apply Loctite® 641 retainer to the shaft where the seal collar (20) will fit. Press the seal collar into position.

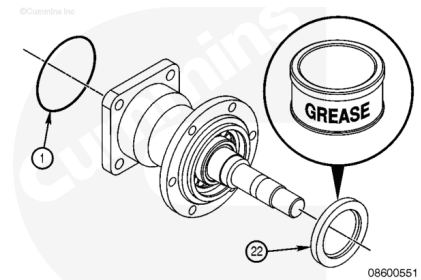
Make sure that excess Loctite™ does **not** enter the bearing.



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Press the lip seal (22) into position. Use grease to lubricate the lips.

Fit the o-ring (1).

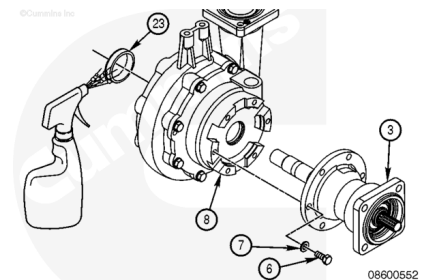


08600551

Press the stationary half of the mechanical seal (23) into the intermediate body (8). Use a 5-percent detergent solution to install the seal.

Secure the intermediate body to the bearing housing with screws (6) and spring washers (7).

Do **not** fully tighten the screws.



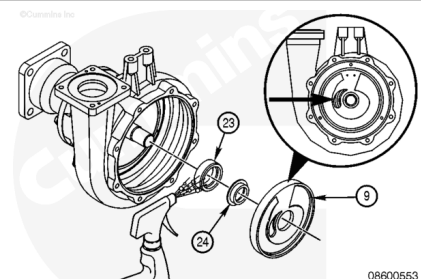
08600552

Use the 5-percent detergent solution to fit the rotating half of the mechanical seal (23) over the shaft.

**Note :** The seal faces must be clean and dry to operate effectively.

Slide the seal spacer (24) over the shaft.

Install the drive end port plate (9) with the port at the 9 o'clock position.

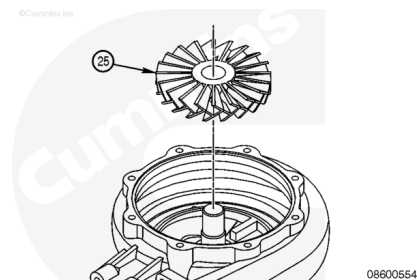


08600553

**Note :** When pressing the priming impeller onto the shaft, make sure that the shaft end is supported to

avoid loading the bearings.

Press the priming impeller (25) onto the shaft.



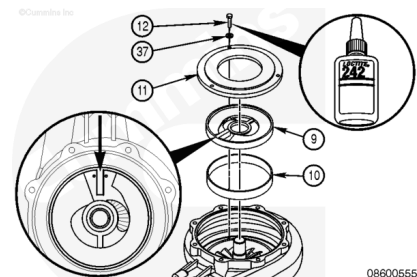
Install the distance ring (10), second port plate (9), and wear ring (11).

**Note :** Make sure the port plate fitting rib is located at the 12 o'clock position so it locates between the two dowels on the wear plate.

Lubricate the screw threads with Loctite® 242 threadlocker.

Install the screws (12) and spring washers (37).

**Torque Value:** 5 n•m [ 44 in-lb ]

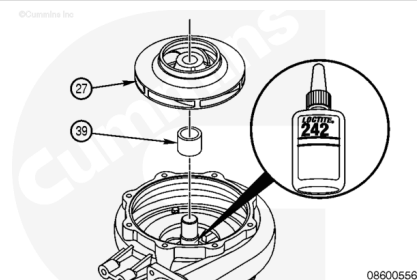


Install the spacer (39).

Screw the impeller (27) onto the shaft using Loctite® 242 threadlocker and a strap wrench.

**Torque Value:** 175 n•m [ 129 ft-lb ]

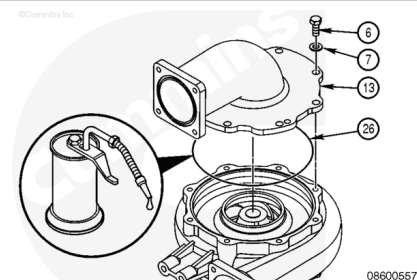
Wipe away the excess Loctite®.



**Note :** When installing the suction bend and pump case on the assembly, it is important that the sealing o-rings are lubricated so as **not** to damage them during assembly.

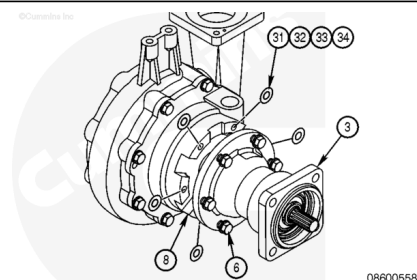
Install a new o-ring (26).

Install the suction bend and pump case assembly (13) using screws (6) and spring washers (7).



The side clearance of the priming impeller **must** be equalized by fitting the appropriate shims (31), (32), (33), and (34) between the bearing housing (3) and the intermediate body (8).

Use a feeler gauge to check the maximum, then the minimum, clearance.



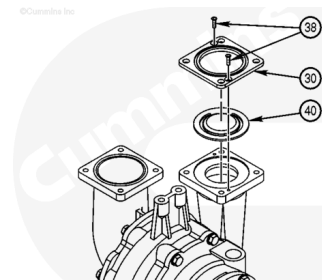
Select the correct shim(s) to equalize the side clearance.  
Subtract the minimum clearance from the maximum clearance. Add one-half of the difference to the minimum to obtain the correct side clearance dimension.

Tighten the screws (6).

**Torque Value:** 25 n•m [ 221 in-lb ]

Install the rubber flap valve (40) into the machined bore. The small plate side fits into the discharge bore.

Install the o-ring flange (30) and secure down with the countersunk screws (38).



## Install

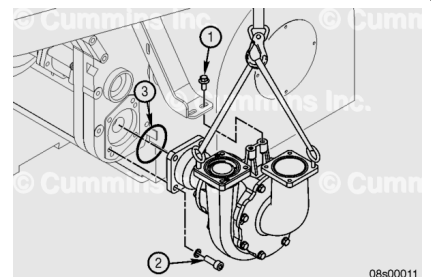
### Gilkes

#### **⚠ WARNING ⚠**

**This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.**

#### **⚠ CAUTION ⚠**

**The sea water pump must be supported by hand or in a sling until the mounting flange is fully engaged with the engine. Do not support the weight of the sea water pump on its shaft end. Damage to the shaft can result.**



Install a new o-ring (3) on the pump mounting flange. Lubricate the o-ring with clean lubricating engine oil.

Use a hoist to lift and support the pump weight, and install the sea water pump.

Install and tighten the four sea water pump mounting capscrews and locking washers (2).

**Torque Value:** 75 n•m [ 55 ft-lb ]

Install and tighten the support bracket capscrews (1).



**Torque Value:** 45 n•m [ 33 ft-lb ]

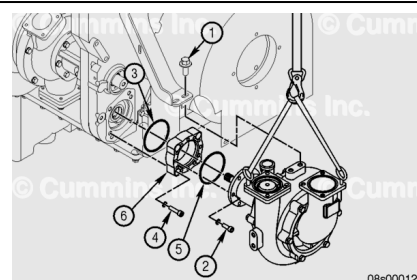
Jabsco®

**⚠ WARNING ⚠**

**This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.**

**⚠ CAUTION ⚠**

**The sea water pump must be supported by hand or in a sling until the mounting flange is fully engaged with the engine. Do not support the weight of the sea water pump on its shaft end. Damage to the shaft can result.**



Install a new o-ring (3) to the mounting adaptor plate (6).  
Lubricate the o-ring with clean lubricating engine oil.

Install the mounting adaptor to the front gear cover.

Install and tighten the four capscrews and washers.

**Torque Value:** 75 n•m [ 55 ft-lb ]

Install a new o-ring (5) on the pump mounting flange. Lubricate the o-ring with clean lubricating engine oil.

Use a hoist to lift and support the pump weight, and install the sea water pump.

Install and tighten the six sea water pump mounting capscrews and washers (2).

**Torque Value:** 45 n•m [ 33 ft-lb ]

Install and tighten the support bracket capscrews (1).

**Torque Value:** 45 n•m [ 33 ft-lb ]

## Finishing Steps

Gilkes

**Note :** The pump must be filled with water before its first use and after being drained for any reason.

- Connect the pipework to the water pump. Make sure that no loads are imposed on the pump.
- Fill the pump with water.
- Open the supply and return valves to the sea water pump.
- Operate the engine and check for leaks.



## Jabsco®

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